Roll No. | | | | | | | | | | | | **Total No. of Pages : 02**

Total No. of Questions : 13

B.Pharma (Sem.–3)

PHARMACEUTICAL ENGINEERING

Subject Code : BP-304T

M.Code : 93326

Date of Examination : 19-01-2023

Time : 3 Hrs. **Max. Marks : 75**

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.**

SECTION-A

1. Explain briefly :

a) What do you mean by pharmaceutical engineering?

b) How can we prevent environmental pollution?

c) What do you mean by Reynolds number?

d) What is the role of manometers?

e) Define heat exchanger.

f) What is size separation?

g) Discuss the disadvantages of the ball mill.

h) Define evaporation.

i) What is molecular distillation?

j) Describe the filtration importance.

SECTION-B

2. Give a note on the Bernoulli theorem.

3. Discuss the complete note on the fluid energy mill with a proper diagram.

4. Write a comment on materials of pharmaceutical plant construction.

SECTION-C

5. Give a detailed note on the horizontal tube evaporator.

6. Discuss in detail about factors affecting size reduction.

7. What are the various parts of a cyclone separator? Discuss it with a diagram.

8. Elaborate about the tray dryer.

9. Discuss the various parts of the silverson emulsifier with a diagram.

10. Give an overview of the demerits of vacuum dryers.

11. Justify the EMC.

12. What is a cartridge filter? Give a complete description.

13. Mention about basics of material handling systems.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.